

AIM:

To write a C program that reads an unsigned integer value, extracts the last four bits individually using the READBIT macro, modifies the most significant four bits of another variable using SETBIT and RESETBIT macros, and displays the resulting output value.

PROCEDURE

1. Start the program.
2. Declare an integer array of size 10 and required variables.
3. Read 10 elements from the user and store them in the array.
4. Traverse the array and count the number of even elements.
5. Sort the array in ascending order using nested for loops.
6. Display the sorted array in ascending order.
7. Sort the array in descending order using nested for loops.
8. Display the sorted array in descending order.
9. Compare array elements and remove duplicate elements by shifting values.
10. Display the array after removing duplicate elements.
11. End the program.

CODE:

```
/*
```

```
File Name : bitman.c
```

```
Author : Monish Kanna
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Date : 12-01-2026
```

```
Purpose : Program to read the last 4 bits of an input value using READBIT macro and  
modify the MSB 4 bits of an output value using SETBIT and RESETBIT macros
```

```
*/
```

```
#include <stdio.h>
```

```
#define READBIT(x, n) ((x >> n) & 1)
```

```
#define SETBIT(x, n) (x |= (1 << n))
```

```
#define RESETBIT(x, n) (x &= ~(1 << n))
```

```
int main()
```

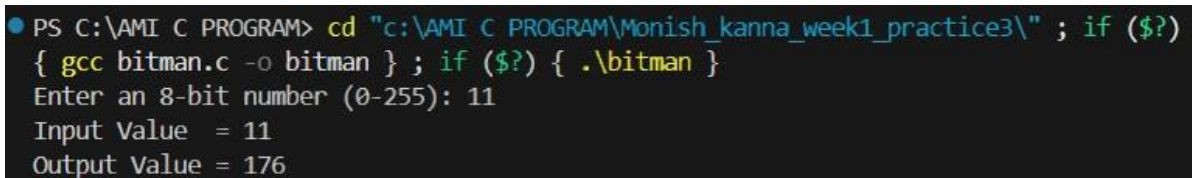
```
{
```

```

unsigned char InputValue;
unsigned char OutputValue = 0;
int bit, i;
printf("Enter an 8-bit number (0-255): ");
scanf("%hhu", &InputValue);
for (i = 0; i < 4; i++)
{
    bit = READBIT(InputValue, i);
    if (bit == 1)
    { SETBIT(OutputValue, i + 4); }
    else
    { RESETBIT(OutputValue, i + 4); }
}
printf("Input Value = %d\n", InputValue);
printf("Output Value = %d\n", OutputValue);
return 0;
}

```

OUTPUT:



```

PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week1_practice3\" ; if ($?) { gcc bitman.c -o bitman } ; if ($?) { .\bitman }
Enter an 8-bit number (0-255): 11
Input Value = 11
Output Value = 176

```

RESULT

The program correctly read the last four bits of the input value and modified the most significant four bits of the output value using bit manipulation macros